Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building responsive web applications, multi-tier REST services, and relational data workflows. Skilled in Java, JavaScript, TypeScript, Angular, Spring Boot, SQL, testing, and Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant application services on Azure SQL and Blob Storage for 500+ users, applying row-level security, normalized schemas, and indexed queries to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant asynchronous workflows for 1,000+ monthly jobs with explicit state transitions, retries, failure isolation, and production monitoring, improving throughput by 3x while preventing cascading service failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered React interfaces and Flask REST services handling 10,000+ daily requests, integrating secure file ingestion, AWS RDS persistence, and modular workflows while maintaining stable performance during peak usage.
- Implemented concurrent processing for 1,000+ uploaded files with validation, error isolation, structured logging, code reviews, and automated tests, reducing feedback turnaround by 60% for 150+ instructors.

USC AD Labs **Los Angeles, CA**
Machine Learning Researcher *January 2025 – May 2025*
- Built Python pipelines across 100,000+ trajectory records, producing reproducible metrics, logs, and visual summaries that improved debugging, benchmarking, and experiment traceability for interactive algorithmic system analysis.
- Documented experimental behavior, edge cases, and validation outputs for researchers, translating complex system signals into actionable findings while maintaining testable procedures and reproducible technical artifacts for team review.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform microservices with object-oriented design, Jenkins pipelines, automated builds, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, strengthening release quality and application maintainability.
- Optimized API and relational data workflows using AWS Glue, Amazon S3, SQL, indexing, caching, and pagination, improving application response times by 30% and production pipeline efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed responsive MERN application workflows with REST APIs and WebSockets, synchronizing 2M+ records across modular services while reducing conflicts through validation controls, fault handling, and real-time state updates.
- Automated Docker-based builds and deployment pipelines using Jenkins CI/CD, unit validation, and release checks, reducing deployment time by 50% while improving repeatability across development and production environments.

Skills

Languages: Java, JavaScript, TypeScript, Python, SQL, C++, Bash
Frontend: Angular, React, HTML5, CSS3, TailwindCSS, responsive UI, state management, component-based design
Backend & APIs: Spring Boot, Spring MVC, REST APIs, Flask, FastAPI, Node.js, Express.js, microservices, OOP
Data & Cloud: PostgreSQL, MySQL, Azure SQL, SQL Server, MongoDB, AWS RDS, Amazon S3, Microsoft Azure
Testing & Delivery: JUnit, PyTest, Mockito, unit testing, TDD, Git, Jenkins, CI/CD, Jira, Agile, MVC, code reviews, debugging

Projects

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven, Node.js*
- Built a full-stack SQL workspace across Java backend and TypeScript frontend modules, supporting 10+ relational and NoSQL databases with schema exploration, validated query execution, and cross-platform deployment workflows.

HMIS | *Java, React, Appwrite, REST APIs, TailwindCSS*
- Developed responsive appointment and patient-history workflows using Java, REST APIs, and authenticated persistence, enabling clinicians to access records 2x faster and coordinate 100+ appointments through user-centered interfaces.

Linguistix | *Python, AWS, Docker, CodePipeline, REST APIs*
- Developed a containerized REST data-processing service with AWS CodePipeline releases, automated validation, structured error handling, and scalable interfaces, achieving 87% accuracy while strengthening repeatable production deployment workflows.